

# Appendix

## A1 Additional Figures and Tables

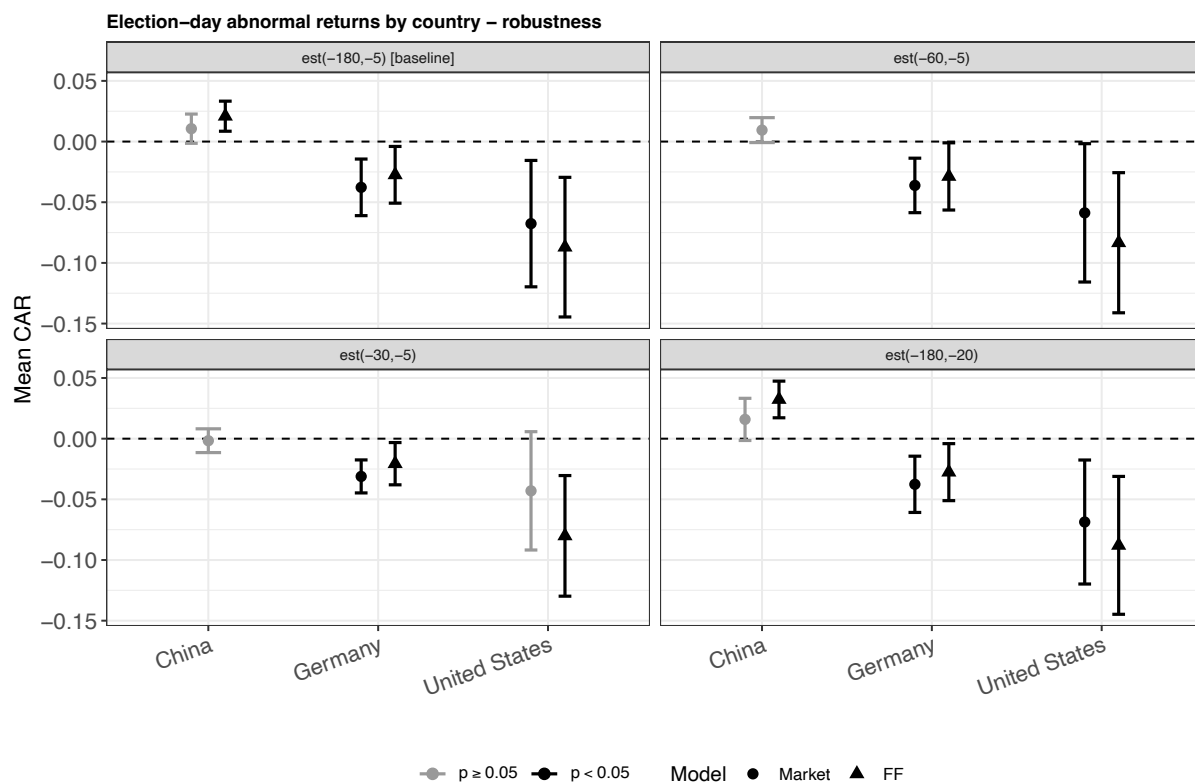
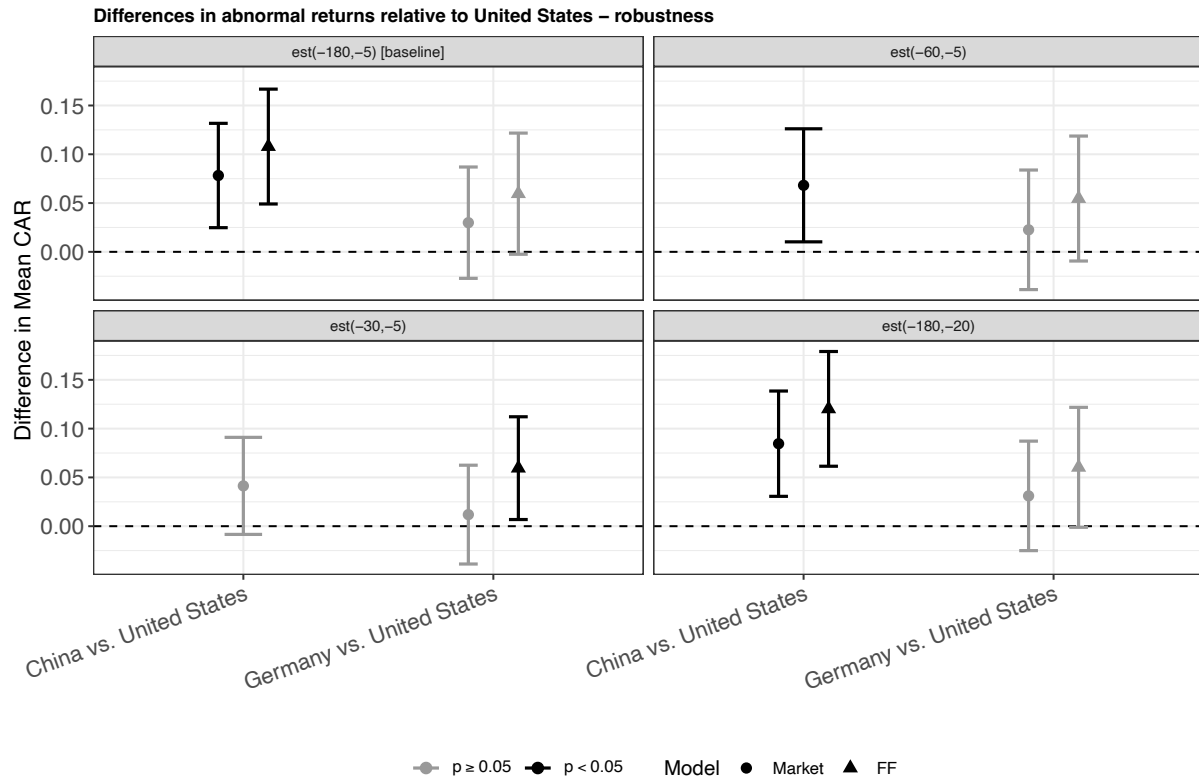


Figure A-1: Robustness to Alternative Estimation Windows, Country-Level CARs



**Figure A-2: Robustness to Alternative Estimation Windows, Differences in Country-Level CARs**

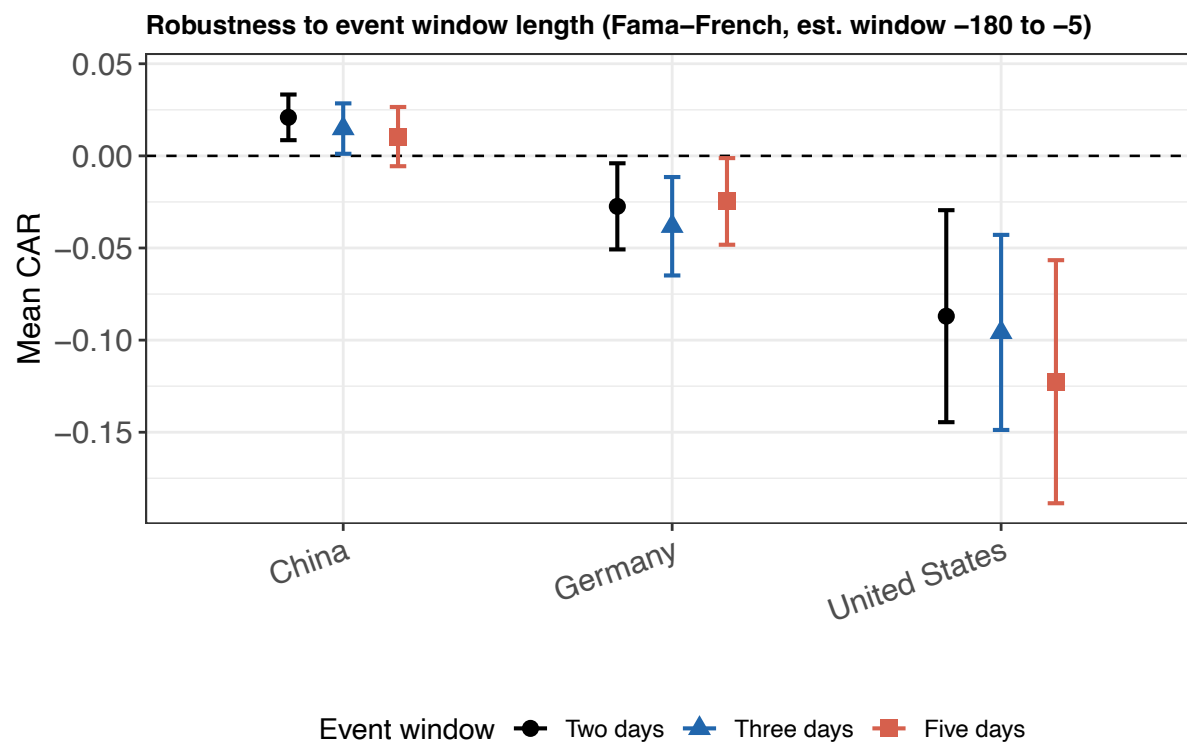


Figure A-3: Robustness to Alternative Event Windows, Country-Level CARs

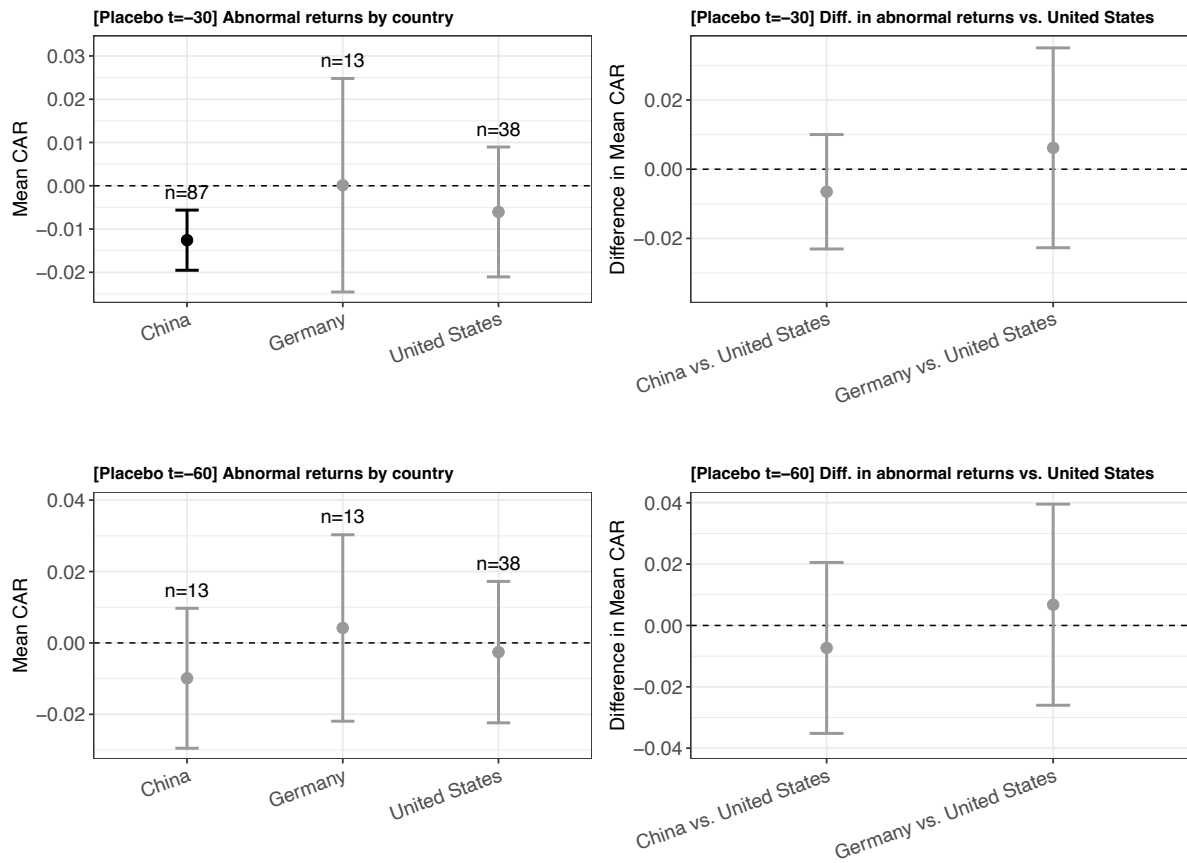


Figure A-4: Placebo Estimation and Event Windows, Country-Level CARs

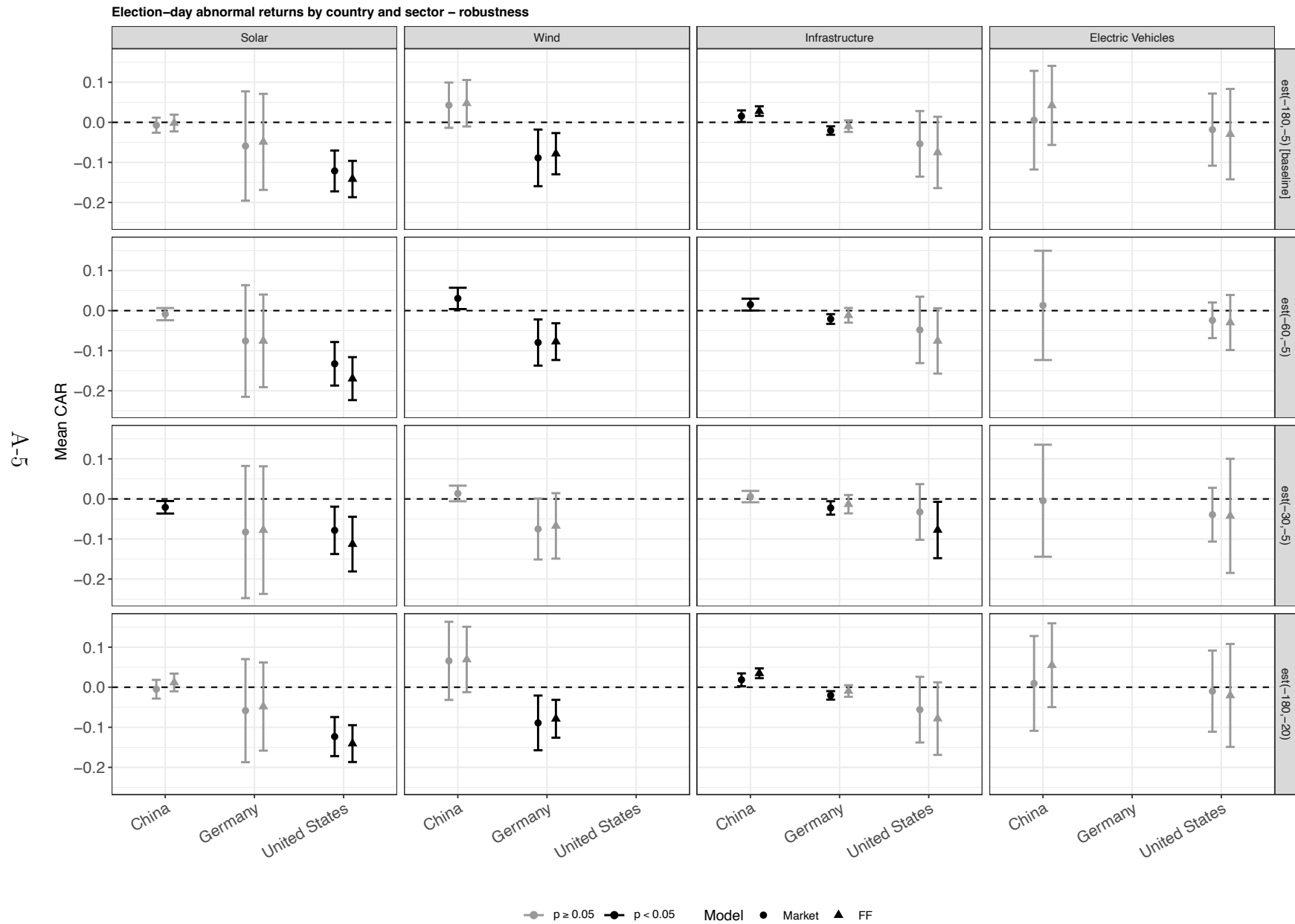
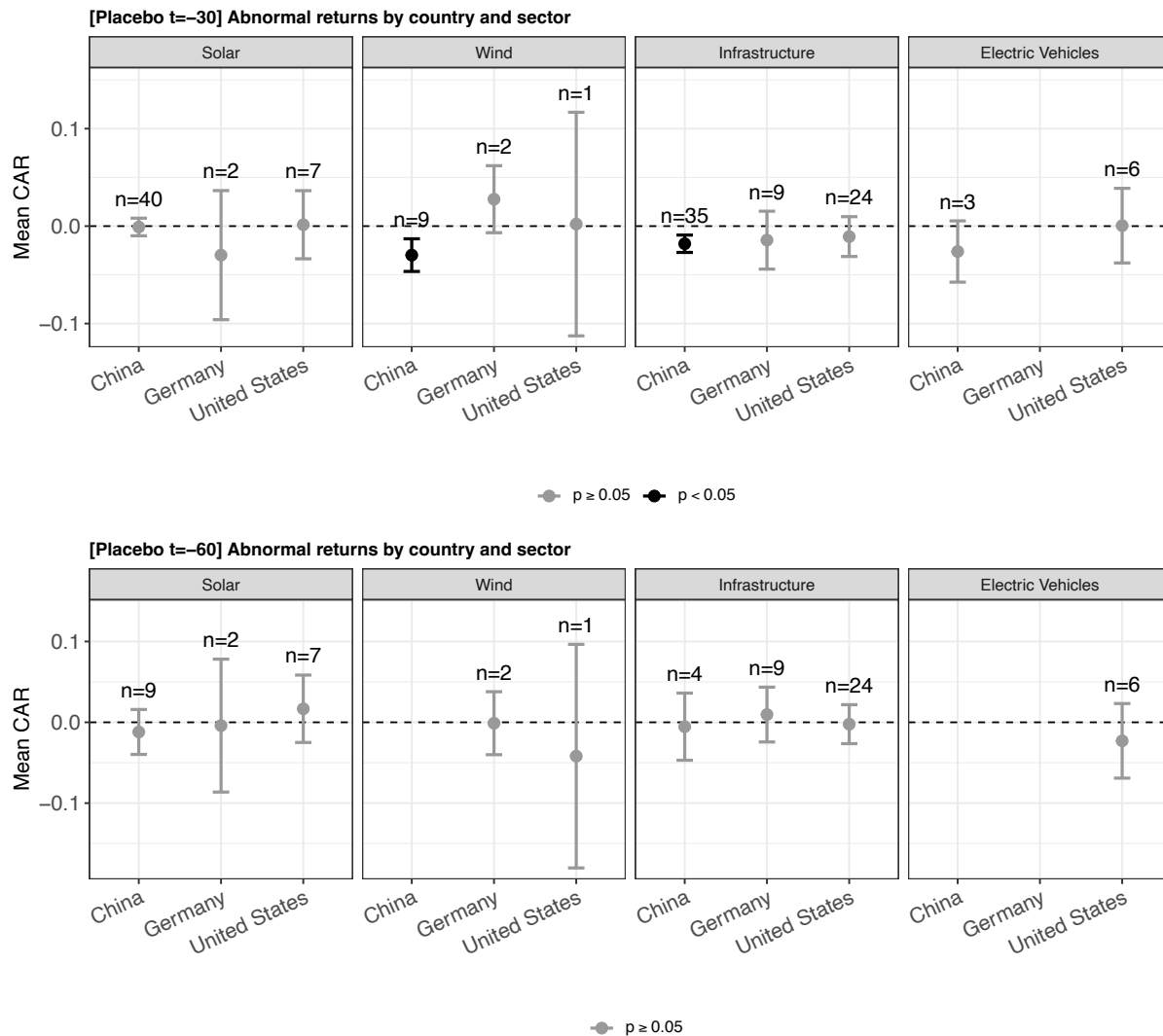


Figure A-5: Robustness to Alternative Estimation Windows, Country-Sector-Level CARs



**Figure A-6: Placebo Estimation and Event Windows, Country-Sector-Level CARs**

**Table A-1: Mean abnormal returns by country**

| Country       | $N$ | Estimate | Std. Error | 95% CI           | $p$   |
|---------------|-----|----------|------------|------------------|-------|
| China         | 85  | 0.021*** | (0.006)    | [0.009, 0.033]   | 0.001 |
| Germany       | 13  | -0.027*  | (0.012)    | [-0.051, -0.004] | 0.022 |
| United States | 49  | -0.087** | (0.029)    | [-0.145, -0.029] | 0.003 |

*Note:* IVW regression, HC3 standard errors. Two day event window. Estimation window  $(-180, -5)$ . Fama-French three-factor model. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-2: Differences in abnormal returns relative to United States**

| Contrast                  | Estimate | Std. Error | 95% CI          | <i>p</i> |
|---------------------------|----------|------------|-----------------|----------|
| China vs. United States   | 0.108*** | (0.030)    | [0.049, 0.167]  | 0.000    |
| Germany vs. United States | 0.060    | (0.032)    | [-0.002, 0.122] | 0.062    |

*Note:* IVW regression, HC3 standard errors. Two day event window. Estimation window  $(-180, -5)$ . Fama-French three-factor model. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-3: Mean abnormal returns by country and sector**

| Sector            | Country       | $N$ | Estimate  | Std. Error | 95% CI           | $p$   |
|-------------------|---------------|-----|-----------|------------|------------------|-------|
| Solar             | China         | 35  | -0.002    | (0.011)    | [-0.023, 0.019]  | 0.871 |
|                   | Germany       | 2   | -0.049    | (0.061)    | [-0.169, 0.071]  | 0.425 |
|                   | United States | 12  | -0.141*** | (0.023)    | [-0.187, -0.096] | 0.000 |
| Wind              | China         | 10  | 0.048     | (0.030)    | [-0.010, 0.106]  | 0.107 |
|                   | Germany       | 2   | -0.078**  | (0.026)    | [-0.130, -0.027] | 0.003 |
| Infrastructure    | China         | 36  | 0.028***  | (0.006)    | [0.016, 0.040]   | 0.000 |
|                   | Germany       | 9   | -0.009    | (0.007)    | [-0.024, 0.005]  | 0.198 |
|                   | United States | 30  | -0.075    | (0.045)    | [-0.164, 0.014]  | 0.098 |
| Electric Vehicles | China         | 4   | 0.042     | (0.050)    | [-0.056, 0.141]  | 0.401 |
|                   | United States | 6   | -0.029    | (0.058)    | [-0.142, 0.083]  | 0.610 |

*Note:* IVW regression, HC3 standard errors. Two day event window. Estimation window  $(-180, -5)$ . Fama-French three-factor model. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-4: Mean abnormal returns by country — robustness to event window length**

|               | <i>Two days</i>     | <i>Three days</i>    | <i>Five days</i>     |
|---------------|---------------------|----------------------|----------------------|
| China         | 0.021***<br>(0.006) | 0.015*<br>(0.007)    | 0.010<br>(0.008)     |
| Germany       | -0.027*<br>(0.012)  | -0.038**<br>(0.014)  | -0.025*<br>(0.012)   |
| United States | -0.087**<br>(0.029) | -0.096***<br>(0.027) | -0.123***<br>(0.034) |

*Note:* IVW regression, HC3 standard errors. Fama-French three-factor model. Estimation window  $(-180, -5)$ . Standard errors in parentheses. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-5: Mean abnormal returns by country — robustness across estimation windows**

|               | <i>est</i> (-180,-5) |                     | <i>est</i> (-60,-5) |                     | <i>est</i> (-30,-5)  |                     | <i>est</i> (-180,-20) |                     |
|---------------|----------------------|---------------------|---------------------|---------------------|----------------------|---------------------|-----------------------|---------------------|
|               | Market               | FF                  | Market              | FF                  | Market               | FF                  | Market                | FF                  |
| China         | 0.011<br>(0.006)     | 0.021***<br>(0.006) | 0.009<br>(0.005)    | —                   | -0.002<br>(0.005)    | —                   | 0.016<br>(0.009)      | 0.032***<br>(0.008) |
| Germany       | -0.038**<br>(0.012)  | -0.027*<br>(0.012)  | -0.036**<br>(0.011) | -0.029*<br>(0.014)  | -0.031***<br>(0.007) | -0.021*<br>(0.009)  | -0.038**<br>(0.012)   | -0.028*<br>(0.012)  |
| United States | -0.068*<br>(0.027)   | -0.087**<br>(0.029) | -0.059*<br>(0.029)  | -0.083**<br>(0.029) | -0.043<br>(0.025)    | -0.080**<br>(0.025) | -0.069**<br>(0.026)   | -0.088**<br>(0.029) |

*Note:* IVW regression, HC3 standard errors. Two day event window. Standard errors in parentheses. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-6: Differences in abnormal returns relative to United States — robustness across estimation windows**

|                           | <i>est</i> (-180,-5) |                     | <i>est</i> (-60,-5) |                  | <i>est</i> (-30,-5) |                   | <i>est</i> (-180,-20) |                     |
|---------------------------|----------------------|---------------------|---------------------|------------------|---------------------|-------------------|-----------------------|---------------------|
|                           | Market               | FF                  | Market              | FF               | Market              | FF                | Market                | FF                  |
| China vs. United States   | 0.078**<br>(0.027)   | 0.108***<br>(0.030) | 0.068*<br>(0.030)   | —                | 0.041<br>(0.025)    | —                 | 0.085**<br>(0.028)    | 0.120***<br>(0.030) |
| Germany vs. United States | 0.030<br>(0.029)     | 0.060<br>(0.032)    | 0.023<br>(0.031)    | 0.055<br>(0.033) | 0.012<br>(0.026)    | 0.059*<br>(0.027) | 0.031<br>(0.029)      | 0.060<br>(0.031)    |

*Note:* IVW regression, HC3 standard errors. Two day event window. Standard errors in parentheses. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-7: Placebo test ( $t = -30$ ): mean abnormal returns by country**

| Country       | $N$ | Estimate  | Std. Error | 95% CI           | $p$   |
|---------------|-----|-----------|------------|------------------|-------|
| China         | 87  | -0.013*** | (0.004)    | [-0.020, -0.006] | 0.000 |
| Germany       | 13  | 0.000     | (0.013)    | [-0.025, 0.025]  | 0.992 |
| United States | 38  | -0.006    | (0.008)    | [-0.021, 0.009]  | 0.429 |

*Note:* IVW regression, HC3 standard errors. Pseudo-event window CAR(-30,-29) (FF model). Estimation window  $(-180, -35)$ . Under the null, all estimates should be approximately zero. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-8: Placebo test ( $t = -60$ ): mean abnormal returns by country**

| Country       | $N$ | Estimate | Std. Error | 95% CI          | $p$   |
|---------------|-----|----------|------------|-----------------|-------|
| China         | 13  | -0.010   | (0.010)    | [-0.030, 0.010] | 0.322 |
| Germany       | 13  | 0.004    | (0.013)    | [-0.022, 0.030] | 0.754 |
| United States | 38  | -0.003   | (0.010)    | [-0.022, 0.017] | 0.798 |

*Note:* IVW regression, HC3 standard errors. Pseudo-event window CAR(-60,-59) (FF model). Estimation window  $(-180, -65)$ . Under the null, all estimates should be approximately zero. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-9: Mean abnormal returns by country and sector — robustness across estimation windows**

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|                                  | <i>est</i> (-180,-5) |                      | <i>est</i> (-60,-5)  |                      | <i>est</i> (-30,-5) |                     | <i>est</i> (-180,-20) |                      |
|----------------------------------|----------------------|----------------------|----------------------|----------------------|---------------------|---------------------|-----------------------|----------------------|
|                                  | Market               | FF                   | Market               | FF                   | Market              | FF                  | Market                | FF                   |
| Solar: China                     | -0.007<br>(0.010)    | -0.002<br>(0.011)    | -0.009<br>(0.008)    | —                    | -0.021**<br>(0.008) | —                   | -0.005<br>(0.012)     | 0.012<br>(0.011)     |
| Wind: China                      | 0.043<br>(0.029)     | 0.048<br>(0.030)     | 0.031*<br>(0.014)    | —                    | 0.014<br>(0.010)    | —                   | 0.066<br>(0.050)      | 0.069<br>(0.042)     |
| Infrastructure: China            | 0.015*<br>(0.007)    | 0.028***<br>(0.006)  | 0.015*<br>(0.008)    | —                    | 0.006<br>(0.007)    | —                   | 0.018*<br>(0.008)     | 0.035***<br>(0.006)  |
| Electric Vehicles: China         | 0.005<br>(0.063)     | 0.042<br>(0.050)     | 0.013<br>(0.070)     | —                    | -0.004<br>(0.071)   | —                   | 0.010<br>(0.060)      | 0.055<br>(0.053)     |
| Solar: Germany                   | -0.059<br>(0.070)    | -0.049<br>(0.061)    | -0.076<br>(0.071)    | -0.075<br>(0.059)    | -0.083<br>(0.084)   | -0.078<br>(0.081)   | -0.058<br>(0.066)     | -0.048<br>(0.056)    |
| Wind: Germany                    | -0.089*<br>(0.036)   | -0.078**<br>(0.026)  | -0.080**<br>(0.029)  | -0.077***<br>(0.023) | -0.075<br>(0.039)   | -0.067<br>(0.042)   | -0.089*<br>(0.035)    | -0.079**<br>(0.024)  |
| Infrastructure: Germany          | -0.020***<br>(0.005) | -0.009<br>(0.007)    | -0.021***<br>(0.006) | -0.012<br>(0.009)    | -0.023**<br>(0.009) | -0.013<br>(0.012)   | -0.020***<br>(0.005)  | -0.009<br>(0.007)    |
| Solar: United States             | -0.121***<br>(0.026) | -0.141***<br>(0.023) | -0.133***<br>(0.028) | -0.170***<br>(0.027) | -0.078**<br>(0.030) | -0.113**<br>(0.035) | -0.123***<br>(0.025)  | -0.141***<br>(0.023) |
| Infrastructure: United States    | -0.054<br>(0.042)    | -0.075<br>(0.045)    | -0.048<br>(0.042)    | -0.076<br>(0.042)    | -0.033<br>(0.035)   | -0.078*<br>(0.036)  | -0.056<br>(0.042)     | -0.078<br>(0.046)    |
| Electric Vehicles: United States | -0.018<br>(0.046)    | -0.029<br>(0.058)    | -0.024<br>(0.023)    | -0.030<br>(0.035)    | -0.039<br>(0.034)   | -0.042<br>(0.073)   | -0.010<br>(0.052)     | -0.021<br>(0.066)    |

*Note:* IVW regression, HC3 standard errors. Two day event window. Standard errors in parentheses. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-10: Placebo test ( $t = -30$ ): mean abnormal returns by country and sector**

| Country                  | $N$ | Estimate  | Std. Error | 95% CI           | $p$   |
|--------------------------|-----|-----------|------------|------------------|-------|
| <i>Solar</i>             |     |           |            |                  |       |
| China                    | 40  | -0.001    | (0.005)    | [-0.010, 0.008]  | 0.847 |
| Germany                  | 2   | -0.030    | (0.034)    | [-0.096, 0.036]  | 0.378 |
| United States            | 7   | 0.001     | (0.018)    | [-0.034, 0.036]  | 0.936 |
| <i>Wind</i>              |     |           |            |                  |       |
| China                    | 9   | -0.030*** | (0.009)    | [-0.047, -0.013] | 0.000 |
| Germany                  | 2   | 0.028     | (0.018)    | [-0.007, 0.062]  | 0.115 |
| United States            | 1   | 0.002     | (0.059)    | [-0.113, 0.117]  | 0.971 |
| <i>Infrastructure</i>    |     |           |            |                  |       |
| China                    | 35  | -0.018*** | (0.005)    | [-0.027, -0.009] | 0.000 |
| Germany                  | 9   | -0.014    | (0.015)    | [-0.044, 0.015]  | 0.342 |
| United States            | 24  | -0.011    | (0.010)    | [-0.031, 0.010]  | 0.303 |
| <i>Electric Vehicles</i> |     |           |            |                  |       |
| China                    | 3   | -0.026    | (0.016)    | [-0.057, 0.005]  | 0.104 |
| United States            | 6   | 0.000     | (0.020)    | [-0.038, 0.039]  | 0.980 |

*Note:* IVW regression, HC3 standard errors. Pseudo-event window CAR(-30,-29) (FF model). Estimation window  $(-180, -35)$ . Under the null, all estimates should be approximately zero. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

**Table A-11: Placebo test ( $t = -60$ ): mean abnormal returns by country and sector**

| Country                  | $N$ | Estimate | Std. Error | 95% CI          | $p$   |
|--------------------------|-----|----------|------------|-----------------|-------|
| <i>Solar</i>             |     |          |            |                 |       |
| China                    | 9   | -0.012   | (0.014)    | [-0.040, 0.016] | 0.400 |
| Germany                  | 2   | -0.004   | (0.042)    | [-0.086, 0.078] | 0.923 |
| United States            | 7   | 0.017    | (0.021)    | [-0.025, 0.058] | 0.431 |
| <i>Wind</i>              |     |          |            |                 |       |
| Germany                  | 2   | -0.001   | (0.020)    | [-0.040, 0.038] | 0.956 |
| United States            | 1   | -0.042   | (0.071)    | [-0.180, 0.096] | 0.553 |
| <i>Infrastructure</i>    |     |          |            |                 |       |
| China                    | 4   | -0.005   | (0.021)    | [-0.047, 0.036] | 0.799 |
| Germany                  | 9   | 0.010    | (0.017)    | [-0.024, 0.044] | 0.579 |
| United States            | 24  | -0.002   | (0.012)    | [-0.026, 0.022] | 0.852 |
| <i>Electric Vehicles</i> |     |          |            |                 |       |
| United States            | 6   | -0.023   | (0.024)    | [-0.069, 0.023] | 0.331 |

*Note:* IVW regression, HC3 standard errors. Pseudo-event window CAR(-60,-59) (FF model). Estimation window  $(-180, -65)$ . Under the null, all estimates should be approximately zero. \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

## A2 Discussion of Sample Construction

We construct our firm sample in two steps. First, we identify publicly traded green firms using Bloomberg’s Industry Classification Standard (BICS), extracting all firms classified under Bloomberg’s renewable energy subsector codes across Chinese, German, and U.S. stock exchanges. Second, we supplement this list with firms identified through a separate Refinitiv/LSEG screen using Thomson Reuters Business Classification (TRBC) codes, retaining any firms present in the LSEG screen but absent from the Bloomberg base. The combined dataset contains 177 firms: 107 listed in China (Shanghai and Shenzhen exchanges), 17 listed in Germany (Frankfurt Stock Exchange), and 53 listed in the United States (NYSE and NASDAQ).

We organize firms into four sectors: Solar, Wind, Infrastructure, and Electric Vehicles. These sectors are constructed from Bloomberg BICS subsector codes and, for firms identified via LSEG, from TRBC codes mapped to their Bloomberg equivalents.

1. **Solar.** The Solar sector includes the full photovoltaic supply chain, from raw feed-stock through balance-of-system equipment. We include firms tagged under the following BICS codes: PV Cells, PV Modules, PV Wafers, Thin Film Modules, Polysilicon, Solar Inverters, Solar Plant Construction, and Semiconductor Materials & Equipment (the last category is retained where Bloomberg applied it to firms whose primary business is photovoltaic manufacturing equipment rather than general semiconductors). From the LSEG supplement, firms tagged under the TRBC code “Photovoltaic Solar Systems & Equipment” are assigned to Solar.
2. **Wind.** The Wind sector includes wind turbine manufacturers and their component supply chains. We include firms tagged under the BICS codes Wind Turbines and Wind Turbine Components. From the LSEG supplement, firms tagged under “Wind Systems & Equipment” are assigned to Wind. One firm, ABO Energy GmbH & Co. KGaA (Germany), is classified by Bloomberg as a “Renewable Electricity” operator, which under our default rules would exclude it; however, Refinitiv/LSEG classifies ABO Energy under “Wind Systems & Equipment,” and its primary business is the development, construction, and sale of wind and solar projects rather than passive electricity operation. We follow the LSEG classification and assign ABO Energy to Wind.
3. **Infrastructure.** The Infrastructure sector covers energy storage, grid modernisation equipment, and related enabling supply chain. We include firms tagged under the BICS codes Energy Storage Systems, Smart Grid Equipment, Electrical Components & Equipment, Heavy Electrical Equipment, Fuel Cells, and Industrial Machinery & Supplies and Construction & Engineering. From the LSEG supplement, firms tagged under “Renewable Energy Equipment & Services (NEC)” and “Waste to Energy Systems & Equipment” are assigned to Infrastructure. Our Infrastructure sector collapses Bloomberg’s Energy Storage and Equipment & Grid categories because the per-cell firm counts for Germany were too small to support separate estimates (two firms in Energy Storage, six in Equipment & Grid).

4. **Electric Vehicles.** The Electric Vehicles sector includes full EV manufacturers and EV-focused supply chain firms. No Bloomberg-base firms carried a BICS code specific to electric vehicles; all EV firms in the sample are drawn from the LSEG supplement, where they are tagged under the TRBC code “Electric (Alternative) Vehicles.”

Several Bloomberg BICS categories present in the raw screen are excluded entirely. Firms tagged as “Renewable Electricity” are excluded because they are electricity generators, operators, or project developers rather than equipment manufacturers or supply chain firms; the theoretical mechanism we study—competitive advantage in manufacturing and technology markets—does not apply in the same way to firms whose revenues derive from selling electricity. Firms tagged under fossil fuel categories (Oil & Gas Refining & Marketing, Coal & Consumable Fuels, Petroleum Retailers, Oil & Gas Storage & Transportation) are excluded as out of scope. We additionally exclude Automotive Parts & Equipment, Marine Transportation, Technology Distributors, Semiconductors, Specialty Chemicals, and Building Products, where Bloomberg’s sector assignment captures firms with no meaningful renewable energy exposure.

Six firms are excluded despite being tagged under in-scope BICS categories, because inspection of their business descriptions reveals that their renewable energy exposure is incidental rather than primary. Table A-12 lists these firms and the rationale for each exclusion.

**Table A-12: Firm-level exclusions**

| Firm                               | Country | BICS category                   | Reason for exclusion                                                                                                      |
|------------------------------------|---------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Riyue Heavy Industry               | China   | Industrial Machinery & Supplies | Generic metal casting and forging; no renewable energy focus evident in business description                              |
| Shandong Shuangyi Technology       | China   | Industrial Machinery & Supplies | Manufactures non-metallic molds across many industries; wind turbine covers are one of many product lines                 |
| Aofu Environmental Technology      | China   | Industrial Machinery & Supplies | Manufactures ceramic products; connection to renewable energy is incidental                                               |
| Norinco International Cooperation  | China   | Construction & Engineering      | Subsidiary of a large state-owned defense/industrial conglomerate; renewable energy construction is a minor business line |
| Guangdong Construction Engineering | China   | Construction & Engineering      | Generic construction conglomerate with broad sector exposure                                                              |
| Runjian Co Ltd                     | China   | Construction & Engineering      | Generic construction conglomerate; primary business is general construction                                               |